

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q21-1859

Axford 138 kV  
September 15, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1859. The Interconnection Customer proposes a 124.9 MW solar photovoltaic facility interconnecting at the Axford 138 kV substation in the Pennsylvania area. The request was received on August 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 634 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,986,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility    | Contingency                     | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------|---------------------------------|-------------|--------------|----------|
| Brent – Axford 138 kV | Orchard Rd – Fort Humbug 500 kV | 137.1       | 312          | 48.8     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------|---------------------------------------------|---------------|-----------------------|
| Axford – Brent 138 kV line | Rebuild line with higher-capacity conductor | \$6,025,000   | 50% (\$2,986,000)     |
|                            | Total allocated to project                  |               | \$2,986,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.